

Exploring transport mobility issues and adaptive behavior of women in a developing country

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ABSTRACT

Women encounter a plethora of mobility challenges in developing countries due to infrastructure challenges and conservative societal norms. This study examines the transport challenges for women across all income groups in a developing country. The data was collected from Dhaka, Bangladesh. It was collected in two stages, using three focus groups and a Revealed Preference travel survey with 2000 women. The focus group discussion revealed that women face challenges in mobility based on their income groups. Social norms and taboos impact all income groups. Among all the income groups, low-income female travelers are the most vulnerable. They expressed that they seem to be a burden to the society and are almost invisible. The findings of this study can contribute to the understanding of women's transport mobility issues in a developing country across all income groups. This study directly contributes to Bangladesh's ability to achieve the Sustainable Development Goals by 2030.

Introduction

Women and girls represent half of the population of the world. Therefore, for a sustainable future, it is essential to end all discrimination towards women and girls and empower them in every respect. The present study provides a gender focus on transport mobility challenges in a developing country. Inefficient and unreliable urban transport systems and inadequate walking facilities significantly restrict women's mobility. Additionally, it is common for women in developing countries, such as Bangladesh, to encounter confrontational and non-confrontational harassment on their daily commute (Asian Development Bank, 2014). Confrontational harassment can take the form of groping, name-calling, and unwanted physical contact, while non-confrontational harassment includes whistling and staring. Women from low-to-middle-income households have limited affordability in choosing their modes of transportation. Despite experiencing harassment, women cannot alter their modes due to a lack of financial affordability (King et al., 2021). This lack of affordable modes restricts their mobility and forces them to avail themselves of only local employment opportunities (Asian Development Bank, 2015).

Although some studies have explored safety and transport mobility issues encountered by women in developing countries, only a limited number of studies explore the differences and similarities in their

transport choices across all income groups. The present study aims to fill this knowledge gap. According to a study by the Transportation Research Board in the USA, dense development, diversified income, reliance on non-polluting modes (such as walking), roads with poor driving surfaces, heavy traffic congestion, uncoordinated institutional structure, and lack of legal capacity to enforce law and regulations are some of the characteristics of a megacity in a developing country (Transportation Research Board, 1996). Being a megacity and capital of Bangladesh, Dhaka suffers from all of these issues. Because of this, Dhaka was selected as the case study city for this research. Findings from this research can be applied to cities with similar characteristics in other developing countries.

All member States of the United Nations (UN) are obliged to accomplish the 17 integrated Sustainable Development Goals (SDGs). As one of the member countries, Bangladesh has to attain all goals, including Goal 1: No Poverty, Goal 5: Gender Equality, and Goal 11: Sustainable Cities and Communities. The target of Goal 1 is to eradicate extreme poverty for all people everywhere by 2030. To eradicate extreme poverty, all men and women, particularly the poor and the vulnerable, should have equal access to economic resources and basic services. The target of Goal 5 is to end all forms of discrimination against all women and girls everywhere and eliminate all forms of violence against all women and girls in the public and private spheres. Adopting

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and strengthening sound policies and enforceable legislation for the promotion of gender equality and the empowerment of all women also has to be ascertained for Goal 5. The target of Goal 11 is to provide safe, affordable, accessible, and sustainable transport systems by expanding public transport, with particular attention to the needs of those in vulnerable situations. Fortunately, socio-cultural constraints on women's mobility in Bangladesh have decreased, with more females getting access to paid work (Babinard, 2011). However, transport mobility constraints still hamper their mobility independence on roads. Without implementing a sustainable transport system, it will be difficult for Bangladesh to achieve Goals 1, 5, and 11 because of women's mobility challenges. A sound transportation system will enable mobility for women travelers, provide ease of access to different services, and reduce poverty through their economic and financial empowerment.

The primary aim of this research is to explore travel behavior across all income categories of women. The scope of the paper is limited to analyzing overall travel behavior, but it does not focus on any specific purpose of the trip. The research questions that this paper will address are:

- 1) What are the mobility issues of female commuters from all income groups?
- 2) How are women across all income groups adapting and meeting their mobility needs?

Data for the present study is collected using two different methods, Focus Group Discussion (FGD) (qualitative data) of three income categories (low, middle, and high) and Revealed Preference (RP) (quantitative data) (i.e., actual travel behavior of female commuters) travel survey.

The following section summarizes relevant literature on transport barriers for women in developing countries and its relevance to gender theory and transport research with Multinomial Logit Model (MNL) analysis. Section "Study Area -Dhaka, Bangladesh" provides background about the city where the case study was done, i.e., Dhaka. Section "Material and Method" provides material and method. Section "Results and Discussions" provides the results and discusses the findings. Lastly, Section "Conclusion" concludes the study with recommendations, limitations, and future directions to achieve the SDG by 2030.

Literature review

Transport barriers for women

In developing countries

In developing countries, travel pattern for women and men varies greatly by their socio-demographic and trip characteristics (Pourhashem et al., 2022; Jain and Geetam, 2020; Basaric et al., 2016; Rosenbloom and Plessis-Fraissard, 2006). The perception of women for transport also varies from men, such as women prefer modes that are less crowded, affordable, and flexible and those that provide door-to-door services such as cycle rickshaws, auto rickshaws, and taxis (Allen, 2018). However, few studies have explicitly focused on the travel behavior of women (Gonzalez and Karla, 2020; Uteng, 2011). As a result, in most developing countries, the transport system continues to be gender-biased (Uteng, 2011).

A household's income contributes significantly to women's travel behavior between social classes (Jain and Geetam, 2020; Mahadevia, 2015; Uteng, 2011). Women's travel is given less priority than men's, with limited access to private and public transport (Lecompte and Bocarejo, 2017; Bhide et al., 2016). In India, this travel difference is also evident due to caste and religion (Mahadevia, 2015). Due to limited affordability, low-income women meet their mobility needs through walking trips (Lecompte and Bocarejo, 2017; Elias et al., 2008; Rosenbloom and Plessis-Fraissard, 2006). With increased income, women can spend more on transport and public transport (Elias et al., 2008;

Rosenbloom and Plessis-Fraissard, 2006). Similarly, Saigal et al. (2021) found that in India, with an improvement in socio-economic status, women switched over to public transport from walking, and men continued to ride motorized vehicles. In developing countries, low- and middle-income women are less likely to own a vehicle or have a driving license. However, in some countries, such as India, women from middle-income backgrounds own two-wheelers, such as motorbikes (Mahadevia, 2015).

Previous research showed that women travel shorter distances to work than men (Lecompte & Bocarejo, 2017; Peters, 2013; Rosenbloom and Plessis-Fraissard, 2006). According to Thynell (2016), low-income women are more inhibited by distance. The issues of income, distance, and family responsibilities are interrelated and significantly impact women's mobility. Irrespective of their income, employed women typically have complex travel patterns (Camstra, 1996; Hanson, 1996). In many Southeast cultures, women carry out most household chores and family responsibilities (Tacoli and Satterthwaite, 2013). Therefore, to reduce travel time and cost, women work closer to home than their male counterparts, even if this means they have to decline a higher-paid job or a better working environment (Arman et al., 2018). On the other hand, limited job opportunities and gender bias in wage rates prevent them from taking a better job opportunity at a further distance, and instead, they choose lower-paid local opportunities (Asian Development Bank, 2010; Hanson, 1996; Hanson & Pratt, 1995). Therefore, women in developing countries face more mobility issues than men due to their social responsibilities and financial restrictions (Asian Development Bank, 2010).

In Delhi, Chennai, Kathmandu, and Karachi, women feel unsafe and are afraid of being physically harassed in crowded public transport (Asian Development Bank, 2014; Jagori, 2010; Mitra-Sarkar and Partheeban, 2009; Paudel, 2011). Kacharo et al. (2022) identified that in Ethiopia, the types of violence in public transport are physical, sexual, and psychological. Their study also indicated that violence against women and girls in Ethiopia in public transport seriously threatens women's safety and security (Kacharo et al., 2022). In Malaysia and Jakarta, the perception of safety is negative for most women compared to men because of spatial configuration and socio-cultural constructs (Hidayati et al., 2020).

Most of the time, women in developing countries cannot defend themselves from misbehavior or physical touching on public transport because of their cultural attitudes and natural inclination (Mannan and Ahmed, 2014). Paudel (2011) identified that women in Nepal protect themselves from sexual harassment on public transport by traveling during the day or avoiding traveling alone. Furthermore, the design of transport-related services, infrastructure, and spaces fails to incorporate safety features such as proper lighting, easy access to public transport, and visibility that could mitigate the risk of assaults on women (Asian Development Bank, 2013). Some metropolitan cities, such as Mexico City, Cairo, Tehran, and Dhaka, offer women-only services or 'pink solutions' and provide buses or metro cars exclusively for women and children (Thynell, 2016). However, Peters (2013) argued that providing separate services can only offer a short-term solution to the problem, but it cannot address the deep-rooted gender bias in transport planning and policymaking.

In Bangladesh

Like other developing countries, harassment is a significant issue faced by women in Bangladesh. However, very limited studies have been conducted that exclusively analyzed the travel issues of Bangladeshi women. Nasrin and Bunker (2021) explicitly mentioned that uncontrolled fares, poor road conditions, lack of comfort, high cost, lack of availability, onboard overcrowding, and lack of safety have to be improved across all modes to improve the transport mobility of women in Bangladesh. Rahman (2022) mentioned that from the perception of low-income women, security of goods, security of passengers, movement flexibility on any road, travel expenditure compared to other

Table 1
Formal and Informal Transport Modes in Bangladesh.

Name of Modes	Description	Types of Modes
Bus	Large buses generally have over 32 seats (10 m or longer). Mini-buses generally have 15 to 30 seats (8 m long).	¹ Formal Public Transport
Auto rickshaw (local name CNG, as driven by compressed natural gas)	Three-wheeled mechanically operated vehicles give door-to-door service to the passengers. Passenger carrying capacity is three persons at the back and the driver in the front.	² Informal Public Transport
Human Hauler (Local name Leguna or Mini-vans)	Human haulers are usually 15 seated in a converted pickup truck with two benches added for passenger seating. Recently, the Government is not providing any route permits to human hauler.	Informal Public Transport
Rickshaw	Three-wheeled cycles, manually operated door-to-door service, carrying capacity is two passengers. Even though the Government decided to limit the number of rickshaws to 87,000, an unofficial estimate claimed that the number of rickshaws plying in Dhaka is about 500,000.	Informal Public Transport
Ride-hailing Service (Shared Taxi, Shared Motorcycle)	Unlicensed taxi service refers to the process of “hailing” or “hiring” a personal driver by a rider to reach the exact location where she or he wishes to go.	Informal Public Transport
Car		Private Transport

¹ Formal Public Transport: High-passenger occupancy vehicles are usually provided by government and other government and non-governmental agencies at users' costs in fares.

² Informal Public Transport: Unconventional or Para transit, not high passenger occupancy vehicles.

transport, and professional attitude of driver are some of the important factors of para transit.

King et al. (2021) stated that in Bangladesh, women faced multiple threats to their safety, including gender-based violence, harassment and crime, and traffic and non-traffic-related injuries while using public transport. Therefore, using public transport is associated with high anxiety and stress levels. A study by the Bangladesh Rural Advancement Committee (BRAC) revealed that 94 % of women who commute by public transport had experienced sexual harassment in verbal, physical, and other forms, while 26 % were harassed on streets, and 82 % experienced leering by men (BRAC, 2018). Lack of social justice is one of the factors identified for the harassment women face on the bus (Islam, Islam, and Haque, 2016). Due to the absence of adequate laws to protect them in public spaces, women tend to adjust their behavior and travel pattern (Hoor-Ul-Ain, 2020). To protect themselves, 21 % of women commuters stopped using public transport, while 45 % started wearing hijab (Muslim headscarf) to appear less attractive and less approachable (BRAC, 2018). Yasir et al. (2022) recommended the introduction of female-only rides and the establishment of small and medium business activities around transit stops to prevent harassment of women.

Transport mobility and feminist theories

The literature review discussed in 2.1 shows that women's travel issues differ from their male counterparts. Even though the travel patterns and needs of female travelers are different, in most developing countries, transport planning always takes the traditional approach of overlooking women's needs. Referring to gender theory, Joelsson and Scholten (2019) mentioned that transport planning research must

acknowledge the diversity of social positions when planning for equity and accessibility. The disparate mobilities of men and women have exposed the gendered conditions of daily life, and feminist and critical epistemologies have given rise to new perspectives on transportation (Joelsson and Scholten, 2019).

The transportation sector is one of the industries that has long been considered a “no-go zone” for women. The “no-go-zone” characteristic of the transportation sector defines Liberal feminism in society, where women do not get equal opportunities as men. The low percentage of female decision-makers in the transportation sector is one of the main reasons for gender bias in the transportation sector (Turnbull, 2013). Moreover, because of the absence of data and evidence on gender-based mobility needs, most of the investment in the transportation sector is male-centric (Legovini et al., 2022).

Violence and harassment toward females define radical feminism in the transportation sector and characterize the Psychoanalytical feminist theory, where the perception of inherited male supremacy justifies the dominating behavior of males toward females. In the public transport system, male dominance is prevalent in the form of harassment towards females while walking or traveling. Because of shame and anxiety, females are afraid to protest against such harassment. Females are afraid because they believe they are fundamentally weak (Sumbal, 2022).

The preferences of male travelers always undermine the transport needs of female travelers, including a lack of emphasis on complex travel behavior (for example, the need for a combination of different trip purposes), a lack of concern for the safety and security of female travelers, and absence of female friendly vehicle design (Rahman, 2022; Bhide et al., 2016). Ignoring these female-specific travel issues in transport planning shows a gender-biased approach. Marxist and Socialist feminist theory explains the deficiency of female needs in the transportation sector.

Sometimes, social and cultural norms for women prohibit using specific modes. In several developing countries, it is typical for women to avoid driving, riding a bicycle or motorbike, and even going out alone, especially at night. It is often the sole responsibility of females to do household work and other duties, such as escorting children to school and grocery shopping. However, the household chores by females are often considered as “doing nothing at home”. Even after engaging in a paid job, females are responsible for the domestic chores. Contrarily, males are privileged to rest after their daily paid job as society automatically assumes that their paid job is more important than that of females, and they are entitled to rest and relax after their day-job is over. In contrast, a female is always “working”, whether for her paid job or for her non-paid domestic chores and responsibilities for her children. Marxist and Socialist feminist theories provide explanations for these social and cultural norms.

To summarize, feminist theories assert that gender is one of the factors of social inequality and conflict. Gender inequality usually benefits men but disadvantages women. Finally, capitalism exploits women by paying low wages and keeping women at home (Macionis, 2010).

Multinomial Logit model (MNL) for travel behavior analysis

McFadden developed the discrete random utility model in 1970 (McFadden, 1973). The Multinomial Logit Model (MNL) is the most well-known application of discrete random utility theory. The Random Utility Models (RUMs) assume that the decision-maker will select one alternative from a finite collection based on the greatest utility (Fischer, 2010; Cascetta, 2009). The mode choice decision is influenced by numerous characteristics of both alternatives and the decision-maker. The MNL has been extensively used to predict mode choice and to analyze travel demand, the relationship between mode choice, and the characteristics of trip makers.

Al-Salih and Esztergár-Kiss (2021) and Mahdi et al. (2022) used logit models to identify which variables have the most significant effect on mode choice for Hungary. Otim et al. (2022) calibrated MNL for

different seasons to predict the transportation mode based on factors such as temperature, air quality, traffic during rush hour, time of day, trip distance, and trip duration in Beijing. Rahman and Baker (2018) developed MNL to assess induced mode switch behavior with the construction of transportation infrastructure in Bangladesh. Meena et al. (2019) calibrated MNL to assess the mode choice for shopping mall trips in Mumbai.

However, there has been no study that employed MNL to understand the mobility issues of female travelers. This study considers several factors to calibrate the MNL to understand the mobility issues of females. A moderate sample size was used to achieve an accurate analysis and a realistic built model. The variables used in the model influence mode choice for females of different income levels. The findings help policymakers and transportation planners to implement a gender-neutral transportation system.

Study area -Dhaka, Bangladesh

Dhaka, the capital of Bangladesh, is an overpopulated city with a population of 8.9 million in an area of 360 km², approximately 16,666 people per km² (Federici, 2004). Due to inefficient transport management, dependency on only road transport, and excessive private vehicles on roads, traffic congestion is a significant problem in Dhaka. The lack of coordination among transport authorities makes the situation even more intolerable (Nasrin, 2015). The descriptions of primary modes of transportation in Dhaka are listed in Table 1. According to Nasrin and Bunker (2021), the pace of walk, rickshaw, leguna (mini-van), CNG, bus, and car are approximately 10 km/hr, 7 km/hr, 4 km/hr, 3 km/hr, 14 km/hr, and 2 km/hr respectively. The cost of walk, rickshaw, leguna, CNG, bus and car are approximately 0 Bangladeshi Taka (BDT)/km (0 USD/km), 1.5BDT/km (0.014 USD/km), 1BDT/km (0.0093 USD/km), 2 km/BDT (0.019 USD/km), 1 BDT/km (0.0093 USD/km), and 300 BDT/km (2.80 USD/km) respectively (Nasrin and Bunker, 2021).

Women comprise 50 % of the population in Bangladesh (Government of Bangladesh, 2019). Around 37 % of the labor force is represented by women (World Bank, 2023; Bangladesh Bureau of Statistics, 2021; Akhter and Islam, 2019). Women are employed in different sectors, such as export-oriented industries, agriculture, manufacturing, and construction. Around 85 % of women are employed in garment factories in Bangladesh, the main economic contributor to Bangladesh (BGMEA, 2014; Islam and Zahid, 2012). A significant percentage of women are involved in informal jobs, such as domestic workers. The household responsibilities, such as household chores and looking after elderly parents and children, lie primarily with women from all income groups.

The Bus Rapid Transit (BRT), Metro Rail, and Elevated Expressway are some of the major projects under construction in Dhaka. Once constructed, the BRT will connect Shahjalal International Airport and Gazipur, located at Dhaka's periphery. The Gazipur area is the garment hub, employing approximately 1 million workers, primarily women. Among all major projects, only the BRT addresses women's issues in design, construction, operation, and management (Asian Development Bank, 2023). Reduced fares for women garment workers, separate queuing systems at the ticket counter for male and female passengers, gender-friendly onboard safety rules, educating drivers about fundamental safety issues, onboard safety awareness announcements, female crews along with male inside buses, lighting facility and underpass near garment factories, police petrol including female police at the foot-over bridges and bus stops, toilets for female passengers, increase participation of women workers in road construction, maintenance, female staffs for BRT management and operation, an equal wage for equal work, availability of water and sanitation facilities for female workers, and provision of shops for women in BRT terminals are some of the issues that have been addressed for women in the gender action plan of BRT (Asian Development Bank, 2023).

Dhaka Elevated Expressway is the first elevated expressway project in Bangladesh. Elevated Expressway will connect the Shahjalal

international airport in Dhaka with Kutubkhali via Mohakhali, Tejgaon, and Kamalapur of Dhaka, Bangladesh. Only one of the six planned metro rails has been constructed that connects Shahjalal airport with Gazipur in the North and Kamalapur in the South. BRT, Metro Rail, and Elevated Expressway will impact the travel pattern of residents in Dhaka. With significant changes occurring in the transport network of Dhaka, it is important to understand barriers encountered by women so that policy, design, and implementation of current and future projects can be impacted in the most effective way.

Material and method

Data collection

As mentioned previously, data collection for this study was undertaken in two stages. The first stage involved Focus Group Discussions (FGD) of females from three income categories. Findings from the FGDs helped in two ways: a) to develop the questionnaire for the second stage RP travel survey, and b) to justify the results from MNL. A participation fee of 200 BDT (about 2 USD) was given to each participant for participating in FGD. The FGD and RP data sets collected from women of different income categories in Dhaka are compatible as they complement each other. FGD provided qualitative data in the form of narratives, opinions, and observations, while RP data focused on quantitative data that was statistically analyzed. The combination of both data sets provided a comprehensive understanding of female transport mobility issues, including rich insights into their experiences, perspectives, attitudes, and significant factors affecting their transport mobility.

Stage 1 focus group Discussions (FGD)

The three FGDs were separated by income level

- Group (1) included low and low-middle-income (0–30,000 BDT per month) female commuters.
- Group 2 included middle-income (30,000 to 60,000 BDT per month) and high-income (above 60,000 BDT per month) female commuters.
- Group 3 involved non-working-female travelers.

The World Bank Atlas (The World Bank, 2023a, 2023b; The World Bank, 2020) was used as an indicator to set the income thresholds. The duration for each FGD was two hours. A detailed discussion was carried out with the participants about their financial situation, transportation usage, problems related to mobility, walking facilities, and ways to improve their mobility options. Participants were recruited through the personal contact of the researchers. Typically, a focus group would consist of 6–10 participants. However, as the topic of women's mobility is broad, a large number (30 in Group 1 and 25 in Group 2 and 3) of participants were recruited. Participants were given equal time to express their ideas and thoughts.

FGD for Group (1) was conducted at the school building of a slum (an informal settlement) located in Mohakhali, Dhaka. This group was divided into two groups, each with 30 participants, one with domestic workers and another with garment factory workers. Separate FGDs were conducted for each group. The age of the participants was between 18 and 50 years.

Discussions with Groups 2 and 3 were held at the University of Asia Pacific campus. The age of participants of both groups was between 25 and 50 years. Private businesses and government agencies employed the majority of participants in Group 2. The primary purpose of travel for participants in Group 3 included shopping and education (education trip as a student and as a mother accompanied by children). The total number of participants in each group was 25. Participants in both groups traveled to their destination by modes such as autorickshaws (CNG), buses, cars, and shared rides (motorbike and automobile).

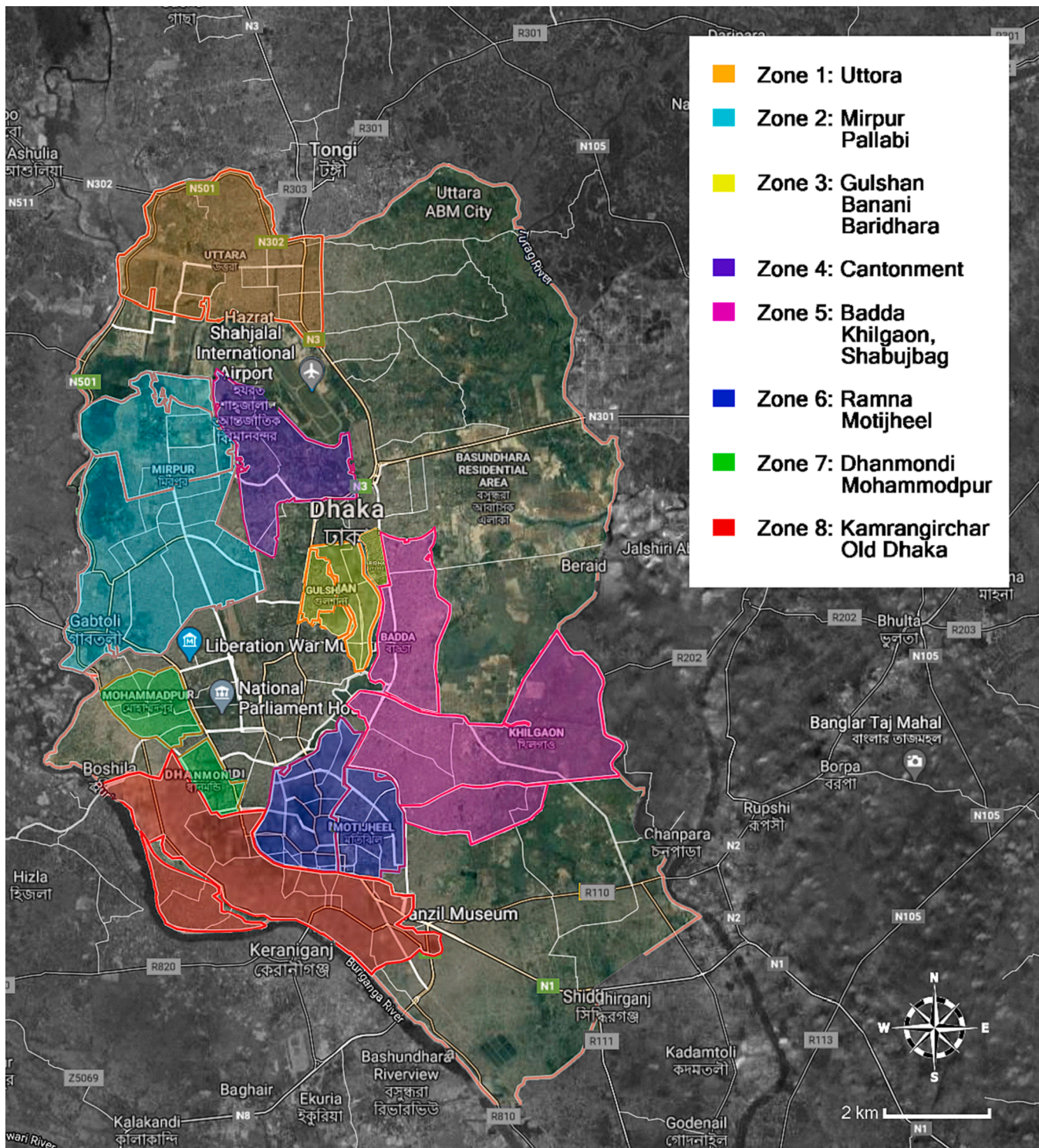


Fig. 1. Locations of Different Zones for RP Travel Survey.

Stage 2 revealed Preference (RP) travel survey

The RP travel survey questionnaire was divided into two sections: questions related to demography and questions on the participants' current travel behavior. Appendix A summarizes the questions included in the travel survey. Using this questionnaire, data was collected between February 2019 and June 2019. The respondents were selected using the following three steps of random cluster sampling:

- In the first step, Dhaka city was divided into eight zones based on the socio-economic and land use characteristics. The reason behind the zonal division was mainly to have a balanced representation of Dhaka's population. Fig. 1 illustrates the locations of different zones for the RP travel survey.
- In the second step, employment centers, shopping centers, education institutions, and informal settlements were selected randomly from each zone.
- In the third step, respondents were selected randomly.

As internet usage was limited and phone survey was expensive, the survey was paper-based with student helpers. Among 2150 requested participants, a total of 2000 women participated in the survey. Each participant took approximately 10 min to fill out the survey. As the survey was conducted face to face with the presence of the student helpers, the response rate was about 98 %. Respondents were not compensated for participating in the survey.

Analytical approach

The analytical approach adopted for this research is the mixed-method approach of qualitative (FGD) and quantitative (Multinomial Logit Model (MNL)) analysis. The mixed-method approach facilitates understanding real-life mobility experiences in various contexts (Gonzalez and Karla, 2020). RP travel survey questionnaire was framed based on the literature review and responses of participants in FGD. FGD provided input for justifying the MNL results as well. MNL was calibrated with the RP travel survey data. Elasticity values were determined from the MNL results to understand the effect of the change of variables. The findings from FGD and MNL results contributed to achieving the objectives of this research.

A model is known as an MNL if more than two alternatives are available. Equation (1) defines the utility function of alternatives in the MNL. The alternative m will be chosen when the utility of that alternative is at the greatest. Nevertheless, because the error term is a random variable, the utility function as a whole also turns into a random variable, making it impossible to compare the utilities of various modes directly. As a result, probabilities of alternatives are computed, and the option with the highest likelihood is typically selected.

$$U_m = i_1 w_1 + i_2 w_2 + \dots + i_n w_n + E_m \quad (1)$$

Where,

U_m = Utility of alternative m

$i_1, i_2, \dots, i_n$ = Attributes

$w_1, w_2, \dots, w_n$ = Weights of the associate attributes

E_m = Error term

The elasticity value was determined to measure the reaction to changing attributes. The relationship between the percentage change in an attribute and the percentage change in the quantity needed is expressed using this unitless metric (Wallis and Schmidt, 2003). Either direct or cross elasticity can exist (Hensher, et al., 2005). Direct elasticity is the percentage change in likelihood of the alternative for each percentage change in one of its variables. Cross-elasticity refers to the probability of other alternative for changing in proportion to a change in those variables. Elasticity values are categorized according to their magnitude; positive or negative indications denote an increase or decrease in the corresponding modes of price fluctuation (Litman, 2013).

Elasticity was determined for this study using the Probability Weighted Sample Enumeration (PWSE) approach. Louviere et al. (2000) recommend the PWSE method over alternative approaches like sample average and naive pooling. According to Hensher et al. (2005), the PWSE approach results in non-uniform cross elasticity since each decision maker is given a distinct weight. The estimated logit function need not pass through the point designated by sample averages if the MNL is nonlinear (Louviere et al., 2000). Because it ignores how each alternative contributes to the chosen outcome, the naive pooling strategy is not recommended.

Results and discussions

Focus group discussion (FGD)

Mobility issues and adaptive behavior of group 1 (Low-income)

Most of the participants in Group 1 (low-income) stated that they had no other alternative to walking. The females on the upper end of the

low-income spectrum could afford leguna for their long-distance travel. Low-income females prefer to live within walking distance (about 1.5 km) of their workplace in low-cost informal settlements rather than living in a better housing community and spending for their commute to work. Participants who work in garment factories sometimes return home after a night shift. They commonly walk in a group or share the cost of a rickshaw or autorickshaw (CNG) (Table 1) on their return trips at night to avoid verbal and physical abuse. The roads inside informal settlements are narrow (almost no spaces among the houses), broken, and dirty. The walking conditions outside of these informal settlements are dilapidated as well.

After day-long overstressed manual work, low-income female commuters have to walk long distances in bad road conditions. As a result, they become exhausted at the end of the day. On top of that, insufficient walking facilities and physical and verbal abuse during commute add to terrible walking experience. Their daily struggles for transport mobility make them depressed and cause them to be non-productive at work. Participants in this group mentioned that it was difficult for them to give their family quality time due to exhaustion at the end of the day.

Low-income females stated they would rather spend money on their family or their child's school travel than their transportation. During the rainy season, walking on waterlogged roads causes foot infection. One of the respondents from the low-income group expressed, "It feels like I do not belong to the system, as I do not have a good walking facility and limited affordability to obtain public transport". Another participant stated, "No one thinks about us; we seem to be the burden of the society". The frustration with transport mobility of low-income female participants reflects Marxist and Socialist feminist theories (Section "Transport Mobility and Feminist Theories"). Low-income females feel excluded from the society. Females in this group accepted their struggle for transport mobility issues and did not demand for their right to transport mobility. Therefore, the transport mobility issues of females in this group can be explained with Psychoanalytical feminist theory (Section "Transport Mobility and Feminist Theories") as well.

Mobility issues and adaptive behavior of group 2 (Middle-and High-Income)

Among the middle-income females of Group 2, most participants traveled short distances (about 1 km) using rickshaws. For long distances (about 6 km), middle-income females commonly traveled by bus. Few also traveled by leguna (Table 1) for long distances. Those who were in the upper spectrum of the middle-income category used CNG (Table 1) for long-distance travel. Some middle-income females use motorbike ride-share. However, due to the social taboo of riding a motorbike with a male stranger, the use of motorbike ride-share is limited. In this group, high-income females travel by car ride-share or private car as a passenger. Even though most high-income females own a car, it is mainly used for their children's education trips or their spouses' work trips. Females from both middle- and high-income usually do not prefer to walk due to bad road conditions.

Participants from middle-income backgrounds discussed that the main reason for their mobility restriction is the unavailability of affordable modes, safety, and harassment. Some other concerns middle-income females complained about were overcrowded and uncomfortable conditions inside buses and leguna, excessive waiting time for bus, lack of bus stops, bad road conditions, and traffic congestion. Despite facing harassment by other passengers and assistants of bus and leguna, they are not able to switch to other modes. This is mainly because of the unavailability of affordable alternative modes or having a constrained travel budget that prevents private car usage. On many roads in Dhaka, rickshaws are restricted and can be used only for short distances; hence, it is not an alternative mode for females in this group. One of the participants expressed her frustration by saying, "I do not feel safe while I travel by bus because incidences of physical and verbal abuse are rampant. However, I do not have any other choice as I must go to work". The females are constantly objectified and are victims of oppressive and misogynistic attitudes. The transport mobility issues of middle-income participants

Table 2
Characteristics of the Survey Data.

Category	Category Ranges	Category Name	Percentages of Survey Data	Data from other sources in Percentages
Age	0–18		24 %	0–15 age group: 27.71 ¹
	19–60		76 %	15–64 age group: 67.14 ¹
	61+		0 %	64 + age group: 5.16 ¹
Income	0–30,000	Low-Income	34 %	-
	30001–60000	Middle-Income	44 %	-
	Above 60,000	High-Income	21 %	-
Marital Status	Married		42 %	63 ² (2016 Statistics)
	Unmarried		58%	37 ² (2016 Statistics)
Trip Purpose	Work		44 %	-
	Education trip as student		37 %	-
	Education trip as mother accompanying children		7 %	-
	Shopping		4 %	-
	Recreational		1 %	-
	Other		6 %	-
Mode with income category	Walk	Low Income	71 %	-
		Middle Income	28 %	-
		High Income	1 %	-
	Rickshaw	Low Income	12 %	-
		Middle Income	44 %	-
		High Income	17 %	-
	Leguna	Low Income	8 %	-
		Middle Income	0 %	-
		High Income	0 %	-
	CNG	Low Income	3 %	-
		Middle Income	3 %	-
		High Income	2 %	-
	Bus	Low Income	26 %	-
		Middle Income	70 %	-
		High Income	30 %	-
	Car	Low Income	0 %	-
		Middle Income	14 %	-
		High Income	27 %	-
	Other ³	Low Income	2 %	-
		Middle Income	2 %	-
		High Income	2 %	-

¹ The World Bank, 2020.

² Akhter & Islam, 2019.

³ Other modes include, ride-share, university bus, office bus.

fall within Radical, Marxist, Socialist, and Psychoanalytical feminist theories (Section “Transport Mobility and Feminist Theories”).

Responses from the High-income females provide evidence that they are more privileged than other income group females. The main mobility issue they stated is traffic congestion. As driving or riding a motorbike is not socially accepted for women in Bangladesh, those who drive a car or ride a bike also mentioned facing verbal and social abuse. Those who travel by ride-share complained about harassment, such as inappropriate staring while using car ride-share and unwanted physical contact while using motorbike-ride-share. One of the respondents mentioned, “*Sometimes I do not feel safe because of Uber driver’s attitude*”. Another mentioned, “*Because I used a ride-share motorbike to go to work, some of my co-workers awkwardly stared at me and said inappropriate things about me*”. Despite the abuse, they do not have the power to challenge the existing social norms. This indicates that even with higher income, high-income females are less empowered than their male counterparts.

Mobility issues and adaptive behavior of group 3- (Not employed in paid Work)

Group 3 participants included those who were not involved in paid employment (homemakers and students). Their main transport issues are congestion, unavailability of affordable modes, and personal safety. Within the group, those who are from high-income mostly travel by private car. Students mentioned they travel by bus and rickshaw. Mothers who escort their children to school primarily travel by rickshaw or CNG. Due to heavy traffic congestion and the unavailability of affordable modes many of them choose not to go back home. Instead, they prefer to wait outside the school. They do not feel safe if their children travel alone.

General mobility issues and adaptive behavior for all income groups

There are a number of mobility issues that were faced by all groups of female commuters. For example, all female travelers mentioned that they do not feel safe and secure when they walk and travel by public transport. Women prefer crowds or roadside vendors for their safety. Participants from all income groups responded that the presence of street vendors gives them the confidence to travel independently, as they act as natural surveillance. However, the presence of disorganized vendors is an impediment. Generally, female commuters complained about the unregulated fare for all modes of public transport. High fare of rickshaw and CNG was an issue across all income-class women. Due to social taboos, females from any income group usually do not ride bicycles or motorbike. Overall, participants said they do not feel independent irrespective of their income status and travel mode.

Revealed Preference travel survey

Description of participants

Participants’ socio-demographic and trip characteristics are given in Table 2. A comparison of survey data with other sources has been conducted to understand if the survey data reflects the characteristics of the overall population. Around 76 % of the participants are between the working age of 19–60 years. The ratio of married and unmarried participants is similar to other resources. Among the participants low-income commuters commonly walk and use leguna, middle-income commuters typically use bus and rickshaw, and high-income commuters use the car.

Multinomial Logit model (MNL)

The LIMDEP software was used for model calibration because it can estimate a wide range of qualitative and dependent variable models (Greene, 1998). The model choice set was considered based on participants’ responses from the FGDs and survey data. A universal choice set

Table 3
Significant variables from the model.

Types of Variables	Attributes	Description	Variable State	Coded Value	Notation of Attributes
Psychological Variable	Convenient	Perception of convenience is a reason to choose a mode.	Not Convenient	0	PCONV
			Convenient	1	
	Comfort	Perception of comfort is a reason to choose a mode.	Not Comfort	0	PCOM
			Comfort	1	
	Safety	Perception of safety is a reason to choose mode.	Safety is not the reason	0	PSAF
			Safety is the reason	1	
	Availability	Perception of availability of mode as a reason to choose a mode	Availability is not the reason	0	PTAV
			Availability is the reason	1	
	Overcrowded	Overcrowding is issue with traveller.	Not Overcrowded	0	TOVERC
			Overcrowded	1	
*Travel Related Problem as Variable	High_Waiting_Time	High waiting time is issue with traveller.	Not High waiting time	0	THWT
			High Waiting Time	1	
	Vendors'_Presence	Presence of vendors is issue with traveller.	Vendor Not Present	0	TVEN
			Vendor Present	1	
	Unavailability	Unavailability is issue with traveller.	Unavailability not a problem	0	TUNAVAI
			Unavailability is a problem	1	
	Uncomfortable	Uncomfortable is the issue with traveller.	Uncomfortable not problem	0	TUNCOM
			Uncomfortable Problem	1	
	No_Alternative	No other alternative mode available is a problem for a chosen mode.	Alternative Available	0	TNAL
			No Alternative	1	
Socio-Demographic Factor	Traffic_Jam	Traffic Jam is a problem.	Traffic Jam is not problem	0	TTJ
			Traffic Jam is problem	1	
	Absence_of_Safety	Absence of safety is problem for a chosen mode.	Mode not safe is not problem	0	TNTSF
			Mode not safe is Problem	1	
	Number_of_Dependent	Number of dependent of a female commuter.			DEP_NUM
	Age_30_Plus	Age above 30 years.	Less than 30	0	A31P
			Above 30	1	
	Income_0_30,000	Income 0 Tk to 30,000 Tk.	Above 30,000	0	IN0_30
			Income 0 to 30,000	1	
	Marital_Status	Marital status of female commuters.	Not Married	0	MAR
Travel Related Variable			Married	1	
	Car_Availability	Availability of car.	Car Available	0	CAR_NO
			No Car Available	1	
	Work_Trip	Trip purpose is work.	Not Work Trip	0	WRK
			Work Trip	1	
	Education_Trip	Trip purpose is education.	Not Education Trip	0	EDU
			Education Trip	1	
	Maximum_Spend	Maximum money female travelers can spend for travel.			MAX_SPEN
	Distance	Distance traveled by female travelers.			DIST
	Travel_Per_Week	Trip frequency per week.			TRAV_WE
	Total_Time	Total travel time			TOT_TIM

*General travel-related problem.

was considered for the model, as most travelers can use any mode for their trip. The choice set for the MNL model includes walk, rickshaw, leguna, CNG, bus, car, and others, mainly shared car and shared motorbike.

According to Ortúzar and Willumsen (2011), the null hypothesis for a significant attribute is rejected when its Student's t statistic is sufficiently large, typically greater than 1.96 at a 95 percent confidence level. Performances of trialied models with different variables were assessed based on their statistical significance, predictive ability, and logical reasoning. Four types of variables were used for model calibration, psychological, travel-related problem, social-demographic, and travel-related variables. After several trials, the variables listed in Table 3 were found to be statistically significant. Except for these variables in Table 3, travel-related problems, harassment, bad road condition, illegal parking, trip-purpose shopping and recreation, and travel cost were used for model calibration. However, none of these variables were significant for the mode choice of females.

Table 4 lists the model estimation results for all statistically

significant variables for the available transport modes. Fig. 2 (a, b, c, d, e) illustrates the direct and cross-elasticity values of walk, rickshaw, leguna, CNG, bus. Appendix B provides direct and cross-elasticity values of different modes with their positive and negative effects.

Effect of Socio-Demographic factors

Results show that socio-demographic variables such as age, income, number of dependents, and marital status significantly impact the mode choice of female travelers (Table 4). The generic variable Age_30_Plus has a positive coefficient for rickshaw, CNG, leguna, and car, which means those who are over 30 years of age have a higher tendency to choose these modes.

The coefficient of variable Number_of_Dependent for a car is negative. In Dhaka, owning several cars is not very common. Participants in FGD stated that household cars are commonly dedicated to education trips for children or male spouses' work trips (Section "Focus Group Discussion (FGD)"). Similarly, the coefficient of the variable

Table 4

Model estimation result for the model calibrated with the survey data.

Mode	Attribute*	Coefficient	Std. Err.	t statistic
Bus	TRAV_WE	0.260	0.058	4.468
	CAR_NO	5.749	1.163	4.945
	PCONV	0.983	0.202	4.861
	TOVERC	-0.701	0.287	-2.445
	PCOM	0.398	0.212	1.873
Rickshaw	DEP_NUM	-0.108	0.052	-2.072
	DIST	-0.281	0.057	-4.957
	INO_30	-0.603	0.224	-2.695
	TRAV_WE	0.137	0.061	2.266
	TVEN	0.794	0.377	2.106
CNG	TUNAVAI	0.908	0.308	2.948
	INO_30	1.269	0.441	2.879
	TRAV_WE	0.208	0.092	2.263
	PTAV	1.889	0.925	2.042
	PSAF	1.644	0.701	2.346
Car	THWT	2.133	0.812	2.629
	INO_30	-1.802	0.852	-2.120
	MAR	0.899	0.415	2.165
	DEP_NUM	-0.245	0.129	-1.903
	CAR_NO	-3.863	0.586	-6.590
Leguna	TRAV_WE	-0.213	0.102	-2.101
	INO_30	2.721	0.856	3.178
	DIST	-0.446	0.164	-2.722
	MAR	-1.165	0.449	-2.593
	DEP_NUM	0.299	0.125	2.389
Walk	MAX_SPEN	-0.001	0.001	-2.650
	TOVERC	0.981	0.562	1.746
	INO_30	1.397	0.254	5.491
	DIST	-0.698	0.079	-8.875
	DEP_NUM	0.162	0.058	2.809
Other	MAX_SPEN	-0.001	0.001	-3.655
	TNAL	0.377	0.197	1.910
	TVEN	1.271	0.427	2.980
	TRAV_WE	-0.380	0.165	-2.304
	A31P	0.387	0.171	2.256
Rickshaw, CNG, Car, Leguna	DIST	0.141	0.051	-3.106
Rickshaw, CNG, Leguna, Walk	CAR_NO	1.56	0.519	3.003
Bus, rickshaw, CNG, Car, Leguna, Walk	TOT_TIM	-0.002	0.002	-1.698
CNG, Leguna	TUNCOM	1.534	0.627	2.446
Car, Leguna, Other	WRK	1.353	0.379	3.567
	EDU	1.339	0.366	3.662
Leguna, Walk	TNTSF	-1.513	0.627	-2.413
Walk, Other	TTJ	-0.462	0.202	-2.291

Log Likelihood Function = 1593 (this value is greater than critical Chi-Square 55.30 at 43 degrees of freedom. So model is significant) (Hensher et al., 2005).

Pseudo $R^2 = 0.43$ (According to (Hensher et al., 2005), this value is equal to linear R^2 of 0.80, which is significant). Sample Size, N = 2000

*Refer to Table 3 for variable codes.

Number_of_Dependent is negative for bus. On the contrary, for leguna and walk, the coefficient of Number_of_Dependent is positive. The cross-elasticity value of walk for the variable Number_of_Dependent is maximum for leguna (Fig. 2 (a), Table B.1), and for leguna it is maximum for CNG (Fig. 2(c), Table B.4). Therefore, it can be said that with the increase in the Number_of_Dependent, most females would switch to walk from leguna. Similarly, for leguna, with the increase in Number_of_Dependent, the maximum percentage of females would switch to leguna from CNG. With the increase in the number of dependents, female travelers who walk or are the primary users of leguna have a higher tendency to save money for their families. Consequently, as leguna costs less and walking is free, the likelihood of choosing leguna and walking increases with the number of dependents.

With the exception of bus, Income_0_30,000 is significant for all modes (Table 4). As the coefficient of Income_0_30,000 is positive for leguna and walk, travelers who are lower-income earners have a higher probability of choosing these modes. Even though participants complained about high fare of CNG in FGD (Section "Focus Group Discussion

(FGD)"), the MNL result showed that the variable Income_0_30,000 is positive for CNG. As low-income females share CNG for traveling to work, the variable Income_0_30,000 is positive for CNG. Leguna is highly elastic (Fig. 2(c), Table B.3), and CNG is relatively elastic (Fig. 2 (d), Table B.4) with Income_0_30,000. Again, the coefficient of Income_0_30,000 is negative for rickshaw, which means those not in low-income will choose rickshaw.

The coefficient of Marital_Status has a positive influence on cars and a negative influence on legunas. As the cross-elasticity of other is maximum and negative, it can be stated that the maximum female travelers would switch to car from other after their marital status change (Fig. 2 (f), Table B.6). Again, cross-elasticity of walk is the maximum for variable Marital_Status change while using leguna (Fig. 2(c), Table B.3). It can be anticipated that those who are married with dual income may have increased affordability to own a private vehicle. On the other hand, travelers who are married and from a low-income background will have a higher tendency to save money for their family, instead of spending money on leguna would choose to walk.

Effect of Traffic-Related variables

Traffic-related variables, include Car_Availability, Travel_Per_Week, Distance, Maximum_Spend, Traffic_Jam, High_Waiting Time, Overcrowded, and Total Time, have a significant impact on the travel behavior of female travelers.

The result (Table 4) shows that with the increased travel frequency per week, the probability of choosing a car decreases, and the probability of choosing bus, rickshaw, and CNG increases. As the coefficient Travel_Per_Week is negative, it can be anticipated that due to the high cost of travel, those who travel more in a week may not choose to travel by car. Again, as stated in the FGD, car is mainly used for child's education trip or male spouse's work trip, females cannot use car for their everyday travel. Again, as the coefficients of variable Travel_Per_Week are positive and the magnitude of the coefficient is about same for bus, rickshaw, and CNG, it can be said that with the increase in travel, the percentages of mode share of those modes would increase. Because of the increase in variable Travel_Per_Week, the maximum number of females would switch from other to rickshaw (Fig. 2 (b), Table B.2). Again, for the increase in variable Travel_Per_Week, the maximum number of females would switch from CNG to bus (Fig. 2(e), Table B.5), and about the same percentages of females would switch from all modes to CNG (Fig. 2(d), Table B.4). Fig. 2 (f) shows that because of the 1 % increase of variable Travel_Per_Week of car, the probability of CNG, rickshaw, walk, and leguna would increase in ascending order.

For walk, rickshaw and leguna, the coefficient of Distance is negative. Therefore, with the increase in travel distance, the probability of choosing rickshaw, leguna and walk decreases. Fig. 2(a) shows that because of the one percent change of variable Distance for walk, maximum percentage of females would shift to Leguna. With the increase in variable Distance of rickshaw, female travelers would switch from rickshaw to other (Fig. 2(b), Table B.2). As rickshaw users are mainly from middle-and high-income, with the increased distance, they can afford to pay for shared-ride (other include shared-ride) for traveling a long distance. Contrarily, as the influence of variable Distance is positive for bus and CNG, commuters will continue to choose CNG and bus if their travel distance increases (Fig. 2(d) and Fig. 2 (e)).

The coefficients of variable Maximum_Spend for walk and leguna are negative, which implies if travelers' ability to spend for transport increases, they will not choose these modes. Fig. 2(a) shows that with one percent change of Maximum_Spend for walk, maximum percentages of females would shift to leguna. For leguna one percent change of Maximum_Spend, the percentage of modal shift to other modes would be about the same except for car (Fig. 2(b)).

Travelers of all modes are sensitive to the variable Total Time. However, from the magnitude of the coefficients, it can be deduced that the modes are not highly sensitive to travel time. As travelers are bound

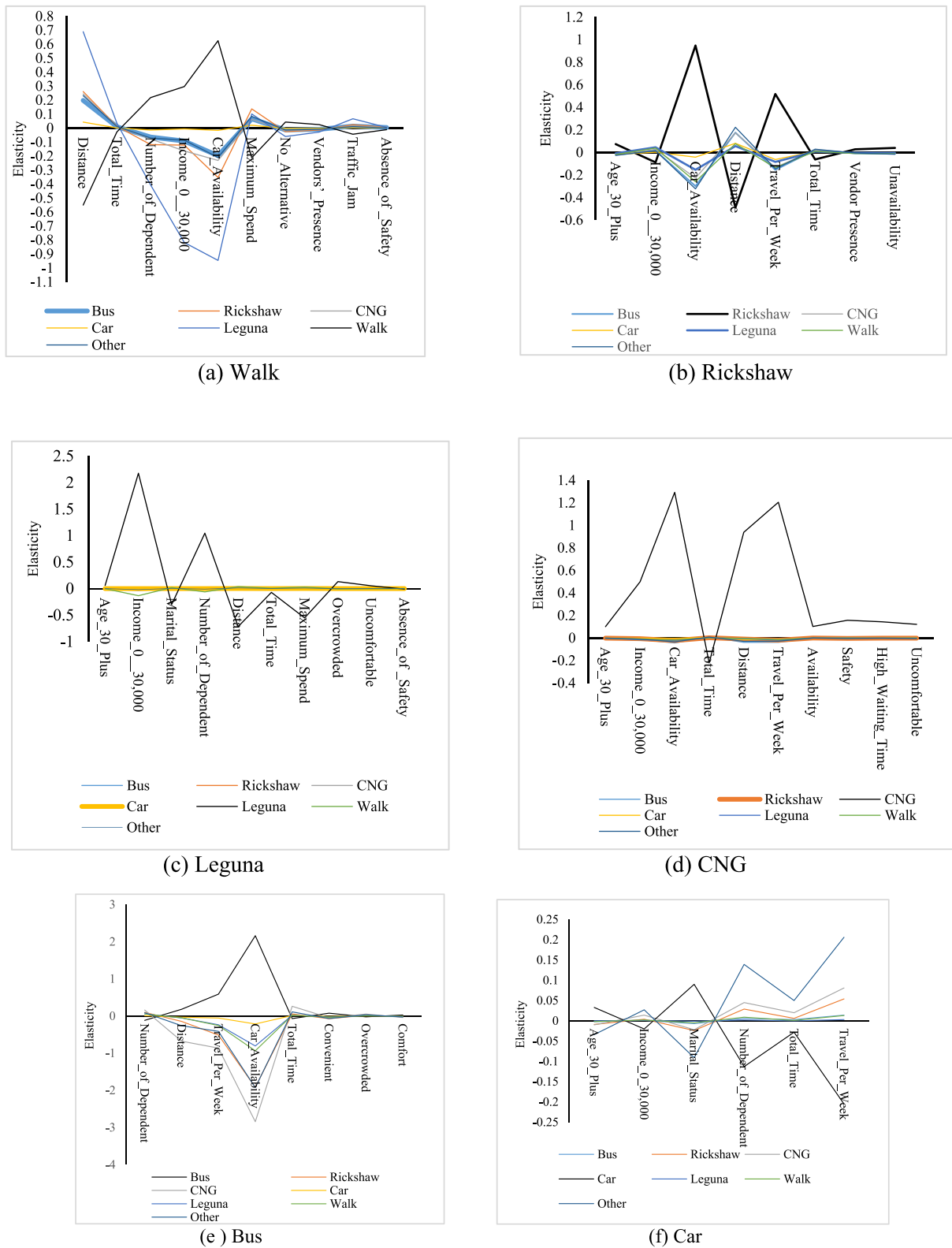


Fig. 2. Direct and Cross-Elasticity of Different Variables for walk, rickshaw, leguna, CNG, and bus. *Black line represents the direct elasticity of the respective mode.

to their respective chosen modes, even if total time increases, travelers cannot necessarily switch modes due to the unavailability of alternate affordable modes.

Effects of Travel-Related problems

Overcrowding has a negative influence on bus. With the increase of variable Overcrowded, bus usage decreases (Table 4). With the increase of bus overcrowding, maximum passengers would switch to CNG

Table A1
Summary of Responses from Travel Survey.

Question	Response Options
Age Range	0–15, 16–18, 19–22, 23–30, 31–40, 41–50, 51–60, 61–70 and 70 plus
Income Range per month	Increment of 5,000 Taka from 0 to 100,000.
Approx. Income of Family	
What is your marital status?	Married, Unmarried
How many dependents do you have?	
Main Travel Purpose	Work, Education as Student, Education as mother accompanying children, Shopping, Recreation, Other
Occupation and Work Type	
How often do you travel in a week?	
How long is your typical journey time?	
Do you have a car available?	Yes, No
Did you ever quit the job because you had no reliable transportation to get to work?	Yes, No
Specify usually which mode do typically use for your main purpose? (Specify Mode, time, cost, waiting time, access time, reason for selecting mode, issues encountered while travelling)	
How much money can you spend for better transportation system?	

(Table B.4). Contrarily, as the coefficient of overcrowding has a positive influence on leguna, therefore, even if leguna is overcrowded female travelers would not switch to any other modes. The cross-elasticity values of other modes are about 0 (Table B.3). This is mainly because low-income women, who are the main users of leguna, cannot afford to pay for higher-cost modes even if leguna is overcrowded.

Coefficients of variable Vendors' Presence are positive for rickshaw and walking, which means that those who use rickshaws or walk perceive the presence of vendors as a travel-related problem. In Dhaka, rickshaw mainly operates inside a locality where roadside vendors occupy half of the roads. Because of the disorganized presence of vendors on roads, it is not easy to walk on roads, and instead, commuters choose to ride rickshaws.

The coefficient of variable Unavailability is positive for rickshaw, which means unavailability of rickshaws is a problem for rickshaw users. As the coefficient of High_Waiting_Time is positive for CNG, high waiting time is a problem for CNG users. As the coefficient of the generic variable Uncomfortable for CNG and leguna is positive, it can be implied that CNG and leguna are uncomfortable for female travelers. Closed and stuffed spaces make CNG and leguna uncomfortable.

The coefficient of the generic variable Absence_of_Safety, is negative for leguna and walk, which means that women do not consider the absence of safety as a problem for them while walking or using leguna. The coefficient of variable Traffic_Jam is negative for walk, which implies that for those who walk, traffic jam is not a problem. As the coefficient of variable No_Alternative is positive, it can be said that the probability to walk is more for those who do not have any other affordable alternatives.

Effects of psychological variables

Psychological variables such as Convenient and Comfort are positive for bus. For one percent change of variable Convenient for bus, females would switch to bus from rickshaw (Fig. 2 (e)). For one percent change of variable Comfort, more female would switch from CNG to bus (Fig. 2 (e)). Even in the FGD, respondents complained about inefficient public transport. Model result (Table 4) shows that bus users consider it Convenient and Comfortable.

Table B1
Direct and Cross-Elasticity Values for Walk.

Mode	Distance	Total_Time	Number_of_Dependent	Income_0_30,000	Car_Availability	Maximum_Spend	No_Alternative	Vendors'_Presence	Traffic_Jam	Absence_of_Safety
Bus	0.198	0.01	-0.064	-0.093	-0.197	0.066	-0.013	-0.008	0.014	0.004
Rickshaw	0.261	0.013	-0.118	-0.122	-0.353	0.138	-0.026	-0.014	0.024	0.006
CNG	0.244	0.012	-0.077	-0.163	-0.231	0.063	-0.018	-0.008	0.014	0.004
Car	0.043	0.002	-0.014	-0.006	-0.016	0.021	-0.002	-0.001	0.003	0.001
Leguna	0.689	0.035	-0.406	-0.818	-0.946	0.101	-0.06	-0.028	0.067	0.005
Walk	-0.55	-0.027	0.218	0.297	0.625	-0.211	0.043	0.025	-0.044	-0.011
Other	0.232	0.012	-0.071	-0.086	-0.21	0.084	-0.014	-0.007	0.011	0.001

*Bold and Italic Values are Direct Elasticity.

Table B2

Direct and Cross-Elasticity Values for Rickshaw.

Mode	Age_30_Plus	Income_0_30,000	Car_Availability	Distance	Travel_Per_Week	Total_Time	Vendor Presence	Unavailability
Bus	-0.021	0.015	-0.299	0.176	-0.155	0.022	-0.007	-0.012
Rickshaw	<i>0.074</i>	<i>-0.088</i>	<i>0.948</i>	<i>-0.489</i>	<i>0.518</i>	<i>-0.062</i>	<i>0.028</i>	<i>0.04</i>
CNG	-0.021	0.028	-0.227	0.173	-0.133	0.022	-0.006	-0.012
Car	-0.017	0.002	-0.042	0.081	-0.063	0.01	-0.005	-0.005
Leguna	-0.008	0.046	-0.154	0.056	-0.084	0.007	-0.004	-0.005
Walk	-0.018	0.039	-0.259	0.077	-0.136	0.01	-0.009	-0.012
Other	-0.026	0.016	-0.323	0.223	-0.146	0.029	-0.005	-0.016

*Bold and Italic Values are Direct Elasticity.

Conclusion

Literature reviews suggest that the travel behavior of men and women varies greatly. However, this is more so in developing countries because of the non-affordability of different transport modes, poor services and infrastructure, and the overburden of social norms and taboos. Dhaka, the capital of Bangladesh, was selected for the case study. The present study examines mobility needs and adapting behavior of women from all income groups in Dhaka. To achieve the objectives of the study, data was collected through FGD and RP travel survey from 2000 respondents.

The current study highlights the necessity of examining the mobility requirements of females from various socio-economic backgrounds. Overall, the MNL results and FGD suggest that women are vulnerable regardless of their social standing and financial situation. The affordability of different modes also differs according to the social standing and income of women.

Feminist theories shed light on the unique mobility issues faced by females of all income and social statuses. However, low-income women are the most vulnerable, as they are often excluded from and overlooked by the transport system despite their direct contribution to the country's economy. They are forced to walk on bad and waterlogged roads because they cannot afford other modes of transportation and live in neighborhoods without access to proper transport facilities. Lack of infrastructure and harassment while walking leaves them feeling like a "burden to the society and like an invisible group in the society," which reflects Radical, Marxist, and Socialist feminist theories. Ironically, even with an increase in income, women's mobility remains restricted. Middle-income females are compelled to use poor-quality public transportation services. While high-income females may be privileged to use cars or ride-share services, social norms prohibit women from driving cars, which shows gender inequality even in the high-income class.

The feminist theory of intersectionality is prevalent in Dhaka. It recognizes that women are not a homogeneous group, and their experiences of oppression vary based on their income class, gender, and other social factors. Low-income women face multiple layers of oppression due to their gender, and income, and they have to rely on natural surveillance to protect themselves from harassment while traveling. Middle-income women, who have more financial resources, may have access to public transportation, but they are still vulnerable to verbal and physical harassment. High-income women, who have more privileges, still face verbal abuse when they drive their cars, which shows that patriarchal norms are deeply embedded in our society. MNL result shows that despite facing harassment, women's mode choice is primarily determined by the availability and affordability of transportation modes. This highlights the importance of addressing the intersectional dimensions of women's mobility issues and providing safe, affordable, and accessible transportation options for all women.

Societal structure and norms reinforce gender-based oppression and limit women's mobility and opportunities. Women who are not engaged in paid jobs and those who are from high-income households face the challenge of being dependent on their family members for transportation, which can limit their mobility and access to opportunities. This reflects the patriarchal attitudes deeply embedded in society that

women should not have access to transportation or be unable to operate a car independently. Marxist and Socialist theories can be applied here.

As most transport policies do not generate the same outcome for men and women, special emphasis should be given to adding a gender action plan in all transport policies. The gender action plan in BRT (Section "Study Area -Dhaka, Bangladesh") can be exemplary for all other infrastructure projects in Bangladesh. The findings of this study can be implemented in the transport policy for Bangladesh, with special emphasis on the following:

- Female travelers face harassment while traveling on the road, despite the model results showing otherwise. Due to limited transportation options, women are forced to choose specific modes and ignore harassment issues while traveling. To address harassment, transport policies should focus on education and awareness about the negative impacts of misbehavior, enforce respect for women, and engage police on walkways and bus stops. Additionally, policies should be in place to organize vendors on sidewalks and encourage business organizations and factory owners to provide transport services for night trips to help female workers reach home safely.
- To improve public transportation for women, steps such as balancing the supply and demand of buses and leguna, deploying more rickshaws to address their unavailability for short distances, ensuring sufficient CNG stands to reduce waiting times, and implementing subsidized public transport for low-income women can be taken. These measures can help women travel safely and comfortably, particularly in areas where access to transportation is limited.
- To promote walking and cycling for short distances, the walkway infrastructure should be enhanced and cycleways built with designs that prioritize comfort. Social media and campaigns should also be used to shift attitudes towards women cycling. Furthermore, CNG and leguna designs should be improved to provide more comfort, while ensuring safe and affordable school transport for children.

Some of the study's limitations include a relatively small sample size and few survey data of shopping and recreation trips. The future scope of the research can be extended into validating the model with a different data set, calibrating different models, and comparing and analyzing the optimum design of different modes for women travelers.

CRedit authorship contribution statement

Sharmin Nasrin: Conceptualization, Methodology, Formal analysis, Validation, Investigation, Data curation, Resources, Writing – original draft, Funding acquisition. **Subeh Chowdhury:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table B3
Direct and Cross-Elasticity Values for Leguna.

Mode	Age_30_Plus	Income_0_30,000	Marital_Status	Number_of_Dependent	Distance	Total_Time	Maximum_Spend	Overcrowded	Uncomfortable	Absence_of_Safety
Bus	-0.001	-0.021	0.003	-0.01	0.011	0.001	0.008	-0.001	0	0
Rickshaw	-0.001	-0.024	0.004	-0.013	0.011	0.001	0.01	-0.001	-0.001	0
CNG	-0.001	-0.041	0.006	-0.018	0.02	0.002	0.011	-0.002	-0.003	0
Car	0	0	0	-0.001	0.001	0	0.001	0	0	0
Leguna	0.059	2.174	-0.303	1.043	-0.717	-0.073	-0.554	0.131	0.052	-0.011
Walk	-0.004	-0.135	0.019	-0.064	0.036	0.004	0.028	-0.009	-0.003	0
Other	0	-0.02	0.002	-0.01	0.011	0.001	0.008	-0.001	0	0

*Bold and Italic Values are Direct Elasticity.

Table B4
Direct and Cross-Elasticity Values for CNG.

Mode	Age_30_Plus	Income_0_30,000	Car_Availability	Total_Time	Distance	Travel_Per_Week	Availability	Safety	High_Waiting_Time	Uncomfortable
Bus	-0.003	-0.015	-0.044	0.009	-0.039	-0.039	-0.004	-0.005	-0.005	-0.004
Rickshaw	-0.002	-0.006	-0.023	0.002	-0.009	-0.02	-0.001	-0.004	-0.002	-0.002
CNG	0.101	0.499	1.293	-0.221	0.938	1.205	0.102	0.157	0.144	0.121
Car	-0.002	-0.002	-0.003	0.004	-0.015	-0.014	-0.002	-0.002	-0.001	-0.001
Leguna	-0.001	-0.016	-0.021	0.001	-0.006	-0.018	0	-0.001	-0.002	-0.002
Walk	-0.001	-0.011	-0.017	0.001	-0.004	-0.014	0	-0.001	-0.002	-0.001
Other	-0.003	-0.01	-0.034	0.008	-0.03	-0.029	0	-0.005	-0.002	-0.005

*Bold and Italic Values are Direct Elasticity.

Table B5

Direct and Cross-Elasticity Values for Bus.

Mode	Number_of_Dependent	Distance	Travel_Per_Week	Car_Availability	Total_Time	Convenient	Overcrowded	Comfort
Bus	-0.117	0.181	0.587	2.161	-0.07	0.077	-0.022	0.027
Rickshaw	0.098	-0.145	-0.513	-1.918	0.056	-0.071	0.021	-0.024
CNG	0.161	-0.677	-0.865	-2.844	0.262	-0.046	0.052	-0.039
Car	0.008	-0.032	-0.055	-0.214	0.013	-0.01	0.003	-0.003
Leguna	0.057	-0.053	-0.234	-0.806	0.021	-0.026	0.006	-0.009
Walk	0.055	-0.05	-0.257	-0.936	0.02	-0.035	0.006	-0.011
Other	0.087	-0.263	-0.414	-1.927	0.111	-0.07	0.033	-0.025

*Bold and Italic Values are Direct Elasticity.

Table B6

Direct and Cross-Elasticity Values for Car.

Mode	Age_30_Plus	Income_0_30,000	Marital_Status	Number_of_Dependent	Total_Time	Travel_Per_Week
Bus	-0.002	0.001	-0.007	0.006	0.003	0.014
Rickshaw	-0.009	0.004	-0.024	0.029	0.006	0.054
CNG	-0.01	0.014	-0.022	0.045	0.02	0.081
Car	0.033	-0.02	0.09	-0.113	-0.028	-0.205
Leguna	0	0.001	-0.001	0.002	0	0.003
Walk	-0.002	0.002	-0.005	0.009	0.001	0.013
Other	-0.034	0.027	-0.091	0.139	0.05	0.206

*Bold and Italic Values are Direct Elasticity.

Data availability

Data will be made available on request.

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Appendix A

Appendix B Direct and Cross-Elasticity Values of Modes

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